

Row V, part 2a: checking your plus-prescription rewrite

Is there a UV divergence? Is the rewrite correct term by term? Can the two log terms go into evolution? Every step is shown and checked numerically.

$$r = z - x, \quad s = y - x, \quad P = x' - w', \quad \xi = \frac{p^+}{k^+}, \quad \bar{\xi} = 1 - \xi = \frac{q^+}{k^+} \text{ with } q^+ = k^+ - p^+, \quad L = \log \frac{k^+ - \Lambda}{\Lambda}, \quad \mathcal{W} = \frac{P \cdot s}{P^2 s^2}$$

Verdict.

1. **No UV divergence: correct.** The reason is the colour factor: it vanishes at $z = x$ (§2). Keep it as one piece; do not split $-ff U(x)U(z)$ from $N_c U(x)$.
2. **The rewrite has one error.** The T_2 term $-\frac{p^+}{k^+ - p^+} [\dots]$ is still in your first part, *and* it is moved to the q^+ part. So it is counted twice, and $\int_0^{k^+}$ of the leftover copy diverges at $p^+ \rightarrow k^+$. **Delete it from the first part.** After that, everything else is right, including signs, prefactors, phases and $\log \frac{k^+ - \Lambda}{\Lambda}$ (§3).
3. **The two log terms are not on the same footing** (§4).
 - The $p^+ \rightarrow 0$ log has a finite z integral. It is the evolution-type term, but its x integral diverges logarithmically at large $|x|$, so it only makes sense after combining with the other rows.
 - The $p^+ \rightarrow k^+$ log has a z integral that **diverges** at large $|z|$. It is not "log $\times$ finite kernel". Done properly, it is a double log, $\frac{1}{2} \mathcal{W} \mathcal{U}_a^\infty L^2$, and should not be put into the evolution kernel.

1. Notation

Put k^+ outside, so that $dp^+ = k^+ d\xi$. Your bracket has six blocks, which I call $T_1 \dots T_6$ in the order written. Each block is contracted with $\frac{P^i r^m}{P^2 r^2}$. The two blocks with poles are

$$T_2 = -\frac{\xi}{\bar{\xi}} B_2, \quad B_2 \equiv \frac{P^i r^m}{P^2 r^2} \left[\frac{(y-x)^i (x-z)^m}{(x-z)^2} - \frac{(y-x)^i (y-z)^m}{2(y-z)^2} \right],$$

$$T_4 = -\frac{\bar{\xi}}{\xi} B_4, \quad B_4 \equiv \frac{P^i r^m}{P^2 r^2} \left[\frac{(y-z)^i (x-z)^m}{(x-z)^2} + \frac{(y-x)^m (y-z)^i}{2(y-x)^2} \right].$$

B_2 and B_4 do not depend on p^+ . The rest of the integrand does:

$$\Phi(p^+) = e^{-ik \cdot (w' - z)} e^{-ik \cdot (z - x) \frac{p^+}{k^+}}, \quad \frac{1}{\mathcal{D}(p^+)} = \frac{1}{p^+(y-x)^2 + (k^+ - p^+)(y-z)^2}.$$

A notation point. Φ depends on p^+ , so it belongs *inside* $\int dp^+$ (and inside $\int_{x,z}$), not in front of the braces. It matters here: the plus prescription subtracts the integrand at the endpoint *including* Φ . That is why your δ terms correctly carry $\Phi(0) = e^{-ik \cdot (w' - z)}$ and $\Phi(k^+) = e^{-ik \cdot (w' - x)}$.

2. Is there a UV divergence? No — step by step

Step 2.1: where a UV divergence could come from. A 2D integral diverges only where the integrand grows like $\frac{1}{(\text{distance})^2}$ or faster. The possible points are $z \rightarrow x$, $z \rightarrow y$ and $y \rightarrow x$. At $z \rightarrow y$ and $y \rightarrow x$, every $\frac{1}{(y-z)^2}$ or $\frac{1}{(y-x)^2}$ comes with one power of the same vector upstairs, so it is only $\sim \frac{1}{\text{distance}}$: finite. The denominator $\mathcal{D} = p^+(y-x)^2 + (k^+ - p^+)(y-z)^2$ vanishes only if $y \rightarrow x$ and $y \rightarrow z$ together. There the measure is $\rho^3 d\rho$ against at most $\frac{1}{\rho^2}$ (after the colour factor below), so that point is finite too. **So only $z \rightarrow x$ ($r \rightarrow 0$) needs checking.**

Step 2.2: the blocks at $r \rightarrow 0$. Several blocks go like $\frac{1}{r^2}$ there. This is exactly the UV behaviour we found in 2b, $\frac{T_{UV}}{s^2} \sim \frac{1}{r^2}$. So without the colour factor, 2a *would* be log divergent.

Step 2.3: the colour factor vanishes at $z = x$. The adjoint Wilson line leaves f^{abc} invariant: $U^{aa'} U^{dd'} U^{cc'} f^{a'd'c'} = f^{adc}$. Multiplying by $U^{db} U^{ce}$ and using $U^T U = 1$ gives $f^{adc} U^{db} U^{ce} = U^{aa'} f^{a'be}$. Contract with f^{fbe} , using $f^{a'be} f^{fbe} = N_c \delta^{a'f}$:

$$f^{adc} f^{fbe} U^{db}(x) U^{ce}(x) = N_c U^{af}(x).$$

Hence your colour factor can be written as

$$-f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x) = -f^{adc} f^{fbe} U^{db}(x) [U^{ce}(z) - U^{ce}(x)].$$

This was checked numerically with $SU(3)$ matrices, to 10^{-15} .

Step 2.4: two powers are gained. $U(z) - U(x) = r \cdot \partial U(x) + O(r^2)$ vanishes linearly. So the integrand becomes $\frac{1}{r^2} \times r \sim \frac{1}{r}$, which is integrable against $d^2 r = \rho d\rho d\phi$. Moreover the linear term is odd in the direction $\hat{r}$, so its angular average vanishes, and what is left is $O(r^0)$.

Check. Take $\rho^2 \times$ the angular average of (integrand $\times$ colour) on a circle of radius ρ around x . A constant would mean a log divergence. For the full 2a integrand it is $4.2 \cdot 10^{-5}$, 10^{-7} , 10^{-9} , 10^{-11} at $\rho = 10^{-2} \dots 10^{-5}$.

For your two new δ terms, the same test gives:

term	$\rho = 10^{-2}$	10^{-3}	10^{-4}
$\delta(p^+)$ term, $\frac{B_4}{(y-z)^2} \times \text{colour}$	$-4.8 \cdot 10^{-4}$	$-4.8 \cdot 10^{-6}$	$-4.8 \cdot 10^{-8}$
$\delta(q^+)$ term, $\frac{B_2}{(y-x)^2} \times \text{colour}$	$-4.7 \cdot 10^{-4}$	$-4.7 \cdot 10^{-6}$	$-4.7 \cdot 10^{-8}$

Both fall to zero, so there is no UV divergence anywhere. **Your statement is correct.**

Caution. This works only because the two colour terms are kept together. Taken separately, $-ffU(x)U(z)$ and $N_c U(x)$ are each UV divergent, with equal and opposite poles.

3. The plus-prescription rewrite, step by step

All constants $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \frac{1}{2\pi}$ and the colour factor are common to every term, so I suppress them until §3.4.

3.1 The $p^+ \rightarrow 0$ pole (block T_4)

(a) Split the coefficient: $\frac{\xi}{\xi} = \frac{k^+ - p^+}{p^+} = \frac{k^+}{p^+} - 1$. So $T_4 = -\frac{k^+}{p^+} B_4 + B_4$. The $+B_4$ is regular, and you have it as $+\dots$. ✓

(b) Let $g(p^+) \equiv \frac{\Phi(p^+)}{\mathcal{D}(p^+)} B_4$. Subtract and add $g(0)$:

$$-\int_{\Lambda}^{k^+-\Lambda} dp^+ \frac{k^+}{p^+} g(p^+) = -\int_0^{k^+} dp^+ \frac{k^+}{p^+} [g(p^+) - g(0)] - k^+ g(0) \int_{\Lambda}^{k^+-\Lambda} \frac{dp^+}{p^+}.$$

- The first term is your $-\left[\frac{k^+}{p^+}\right]_+$. Extending the limits to 0 and k^+ costs only $O(\Lambda)$, because the bracket is finite at both ends.
- In the second, $\int_{\Lambda}^{k^+-\Lambda} \frac{dp^+}{p^+} = \log \frac{k^+-\Lambda}{\Lambda}$ exactly.

(c) The endpoint value: $\Phi(0) = e^{-ik \cdot (w'-z)}$ and $\mathcal{D}(0) = k^+(y-z)^2$, so $k^+ g(0) = \frac{e^{-ik \cdot (w'-z)}}{(y-z)^2} B_4$.

(d) Restore the constants. The δ term is $-\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \frac{1}{2\pi} \log \frac{k^+-\Lambda}{\Lambda} e^{-ik \cdot (w'-z)} \frac{B_4}{(y-z)^2}$. Here $\frac{g^4}{8\pi^4} \cdot \frac{1}{2\pi} = \frac{g^4}{16\pi^5}$, which is exactly your third term. ✓

3.2 The $p^+ \rightarrow k^+$ pole (block T_2)

(a) Split the coefficient: $\frac{\xi}{\xi} = \frac{p^+}{k^+ - p^+} = \frac{k^+}{k^+ - p^+} - 1$. So $T_2 = -\frac{k^+}{k^+ - p^+} B_2 + B_2$.

(b) Change variable, $q^+ = k^+ - p^+$, so $dp^+ = -dq^+$ and $p^+ : 0 \rightarrow k^+$ becomes $q^+ : k^+ \rightarrow 0$. The phase and the denominator become:

- $\Phi = e^{-ik \cdot (w'-z)} e^{-ik \cdot (z-x)} e^{+ik \cdot (z-x) \frac{q^+}{k^+}} = e^{-ik \cdot (w'-x)} e^{-ik \cdot (x-z) \frac{q^+}{k^+}};$
- $\mathcal{D} = (k^+ - q^+)(y-x)^2 + q^+(y-z)^2$.

Both are as you wrote. ✓

(c) With $h(q^+) \equiv \frac{\Phi}{\mathcal{D}} B_2$:

$$\int_{\Lambda}^{k^+-\Lambda} dp^+ \left(-\frac{k^+}{q^+} + 1 \right) h = \int_{\Lambda}^{k^+-\Lambda} dq^+ \left(-\frac{k^+}{q^+} + 1 \right) h = -\int_0^{k^+} dq^+ \left\{ \left[\frac{k^+}{q^+} \right]_+ - 1 \right\} h - k^+ h(0) \log \frac{k^+ - \Lambda}{\Lambda}.$$

Your $\int_{k^+}^0 dq^+ \{ [\frac{k^+}{q^+}]_+ B_2 - B_2 \}$ equals the first term, because reversing the limits gives the minus sign. ✓ (It is clearer to write $-\int_0^{k^+}$, since a plus distribution is defined with the limits in increasing order.)

(d) The endpoint value: $\Phi(q^+=0) = e^{-ik \cdot (w'-x)}$ and $\mathcal{D} = k^+(y-x)^2$. This gives your fourth term, $-\frac{g^4}{16\pi^5} \frac{1}{k^+} \log \frac{k^+-\Lambda}{\Lambda} e^{-ik \cdot (w'-x)} \frac{B_2}{(y-x)^2}$. ✓

3.3 The error: T_2 is counted twice

Steps 3.2(a)–(d) replace T_2 . But your first part still contains $-\frac{p^+}{(k^+-p^+)} \left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2} \right]$, now integrated over $\int_0^{k^+}$.

- It is counted twice.
- Its $\int_0^{k^+} dp^+ \frac{1}{k^+-p^+}$ diverges at $p^+ \rightarrow k^+$.

Check at fixed positions (random configurations, $k^+ = 1$), against the original $\int_{\Lambda}^{k^+-\Lambda}$:

config, Λ	original	corrected rewrite	your rewrite as written
1, 10^{-3}	4.24154 + 0.87153i	4.24045 + 0.87163i	6.14348 + 1.85793i
1, 10^{-5}	7.29054 + 1.45478i	7.29053 + 1.45478i	10.42804 + 3.23198i
2, 10^{-3}	0.36178 + 1.29212i	0.36154 + 1.29240i	1.23186 + 1.93494i
2, 10^{-5}	0.67059 + 2.09737i	0.67059 + 2.09737i	2.32378 + 3.15281i
3, 10^{-3}	34.37460 + 18.01192i	34.36404 + 18.00604i	50.75331 + 30.78769i
3, 10^{-5}	60.56981 + 31.83545i	60.56971 + 31.83539i	89.23602 + 55.00372i

- The corrected rewrite agrees, with a difference that shrinks $\propto \Lambda$.
- Your rewrite as written is off by an amount that grows like $\log \frac{1}{\Lambda}$.

(The "as written" column needs a cutoff at $k^+ - \Lambda$ just to be finite.)

3.4 The corrected formula

In the first part, **delete** the term $-\frac{p^+}{(k^+-p^+)} \left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2} \right]$. Everything else stays as you wrote it. Schematically, with $\mathcal{C}_a = [U^{ab'}(x') - U^{ab'}(w')] [-f^{adc} f^{fbc} U^{db}(x) U^{ce}(z) + N_c U^{af}(x)] \rho^{b'}(x') \rho^f(y)$:

$$\frac{d^3 N_{2a}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_{x,y,z,x',w'} \left\{ \int_0^{k^+} \frac{dp^+}{2\pi} \frac{\Phi(p^+)}{\mathcal{D}(p^+)} [T_1 + T_3 + T_5 + T_6 + B_4] - \int_0^{k^+} \frac{dp^+}{2\pi} \left[\frac{k^+}{p^+} \right]_+ \frac{\Phi(p^+)}{\mathcal{D}(p^+)} B_4 \right. \\ \left. - \int_0^{k^+} \frac{dq^+}{2\pi} \left\{ \left[\frac{k^+}{q^+} \right]_+ - 1 \right\} \frac{\Phi}{\mathcal{D}} \Big|_{p^+=k^+-q^+} B_2 - \frac{\log \frac{k^+-\Lambda}{\Lambda}}{2\pi} \left[e^{-ik \cdot (w'-z)} \frac{B_4}{(y-z)^2} + e^{-ik \cdot (w'-x)} \frac{B_2}{(y-x)^2} \right] \right\} \mathcal{C}_a$$

Here T_1, T_3, T_5, T_6 are your blocks with their p^+ -dependent coefficients, and $\int_0^{k^+} dp^+ \left[\frac{k^+}{p^+} \right]_+ f \equiv \int_0^{k^+} dp^+ \frac{k^+}{p^+} [f(p^+) - f(0)]$.

This identity holds at **fixed** x, y, z, x', w' , up to $O(\Lambda/k^+)$.

4. Do the two log terms "enter the evolution"?

A term $L \times K(x, y, z)$ can be absorbed into an evolution equation only if $\int_{x,y,z} K$ is finite (possibly after combining rows). Check each δ term.

4.1 The $p^+ \rightarrow 0$ log: yes, with one caveat

$$K_0 = e^{-ik \cdot (w'-z)} \frac{B_4}{(y-z)^2} \mathcal{C}_a.$$

- **z integral: finite.** The phase $e^{ik \cdot z}$ oscillates, and at large $|z|$ the term falls like $\frac{1}{|z|^2}$. At $z \rightarrow x$ it is finite (§2).

- **x integral: log divergent at large $|x|$.** The phase $e^{-ik \cdot (w' - z)}$ contains no x . As $U(x) \rightarrow 1$ the colour factor tends to $-f^{abc}f^{bce}U^{ce}(z) + N_c\delta^{af}$, which is not zero (-2.29 in the test projection). And $|x|^2 \langle \frac{B_1}{(y-z)^2} \rangle_{\hat{x}} \rightarrow -\frac{P \cdot (y-z)}{2P^2(y-z)^2}$: numerically 0.396984, 0.396348, 0.396342 at $|x| = 10, 10^2, 10^3$, against the prediction 0.396341. So $\int d^2x \sim \int \frac{d^2x}{x^2}$ diverges.

This is the same large- $|x|$ tail we found for 2b (β_0). It is the usual situation for a soft-gluon log: the tail cancels against the virtual and other real diagrams, and only the sum gives the JIMWLK/BK kernel. **So this log does go into evolution, but only after you add the other Group I rows**, not from 2a alone.

4.2 The $p^+ \rightarrow k^+$ log: no — it is really a double log

$$K_1 = e^{-ik \cdot (w' - x)} \frac{B_2}{(y-x)^2} \mathcal{C}_a.$$

- **z integral: log divergent at large $|z|$.** The phase $e^{-ik \cdot (w' - x)}$ contains no z .
- Outside the target $U(z) \rightarrow 1$, and the colour factor tends to $\mathcal{U}_a^\infty (\propto -f^{adc}f^{fbc}U^{db}(x) + N_c U^{af}(x))$, which is not zero (-1.03 in the test projection).
- From the definition of B_2 : $\frac{B_2}{(y-x)^2} = \frac{\mathcal{W}}{r^2} \left[-1 - \frac{r \cdot (s-r)}{2(s-r)^2} \right] \xrightarrow{|r| \rightarrow \infty} -\frac{\mathcal{W}}{2r^2}$. Numerically, $|z-x|^2 \langle \cdot \rangle = 0.375000$ at $|z-x| = 10, 10^2, 10^3$, against $-\frac{\mathcal{W}}{2} = 0.375000$.
- So $\int d^2z K_1 \sim -\pi \mathcal{W} \mathcal{U}_a^\infty \log R_{\max}$: infinite.

Why. At $q^+ = 0$ exactly, nothing cuts off large z : the phase $e^{-ik \cdot (x-z)q^+/k^+} \rightarrow 1$ and the denominator $\rightarrow k^+(y-x)^2$. For $q^+ > 0$, large z is cut off at $|z-x| \sim |y-x| \sqrt{k^+/q^+}$ by the denominator, and at $|z-x| \sim \frac{k^+}{q^+|k|}$ by the phase. So the z integral and the limit $q^+ \rightarrow 0$ do not commute.

Your $\left[\frac{k^+}{q^+}\right]_+$ term subtracts $h(0)$, and $\int_z h(0)$ is the same divergent integral. So after $\int_z$:

- the $\left[\frac{k^+}{q^+}\right]_+$ term and the $\delta(q^+)$ log term are **separately infinite**;
- only their sum is finite.

Doing $\int_z$ first at fixed q^+ (rowV_full_derivation.pdf, §8), the sum behaves as $\frac{\pi \mathcal{W} \mathcal{U}_a^\infty \log(k^+/q^+)}{q^+/k^+}$ near $q^+ \rightarrow 0$. Half of this comes from T_2 in $|s| < |r| < |s| \sqrt{k^+/q^+}$, and half from T_3, T_4, T_6 further out. After $\int dq^+$ this gives

$$\frac{1}{2} \mathcal{W} \mathcal{U}_a^\infty L^2 + (\text{single } L) + (\text{finite}),$$

not a single L times a finite kernel.

Consequence. Do not put the $\delta(q^+)$ term $-\frac{q^4}{16\pi^5} \frac{1}{k^+} \log \frac{k^+-\Lambda}{\Lambda} e^{-ik \cdot (w'-x)} \frac{B_2}{(y-x)^2}$ into the evolution kernel on its own. Its z integral does not exist. If you want this endpoint in "log $\times$ kernel" form, do the z integral first, keeping $q^+ > 0$. The result is the double log above; it is a Sudakov-type term, not evolution.

5. Summary

item	status
no UV divergence in 2a	correct ; the colour factor $\propto U(z) - U(x)$ removes the $\frac{1}{r^2}$ (keep the two colour terms together)
T_4 : $\frac{\xi}{\xi} = \left[\frac{k^+}{p^+}\right]_+ - 1 + \delta$ term	correct (sign, prefactor $\frac{q^4}{16\pi^5}$, phase $e^{-ik \cdot (w'-z)}$, $\frac{1}{(y-z)^2}$, $\log \frac{k^+-\Lambda}{\Lambda}$)
T_2 : $q^+ = k^+ - p^+$, $\left[\frac{k^+}{q^+}\right]_+ - 1 + \delta$ term	correct (phase, denominator, reversed limits, $\frac{1}{(y-x)^2}$)
T_2 left in the first part	error : counted twice. Delete it.
$p^+ \rightarrow 0$ log into evolution	yes, but its $\int_x$ diverges at large $ x $: combine with the other rows first
$p^+ \rightarrow k^+$ log into evolution	no : its $\int_z$ diverges; done properly it is $\frac{1}{2} \mathcal{W} \mathcal{U}_a^\infty L^2$

Checks: src/rowV_2a_plus_check.py (the pointwise identity with and without the duplicate T_2 , the UV test of the two δ terms with a smooth $SU(3)$ adjoint field, and the large- $|z|$ and large- $|x|$ tails), output in RESULTS_rowV_2a_plus_check.txt. The earlier 2a results come from src/rowV_final.py (step 7) and rowV_full_derivation.pdf (§§7–8).